

Quantum vs. Classical Correlations in Dijet Hadron Multiplicities: A Finite-Statistics Methodology Note

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Abstract

We work through the full methodological chain for measuring mutual information (MI) between back-to-back jets from hadron multiplicity distributions, and for establishing whether an observed correlation could be a genuine quantum signature or is fully explained by classical physics plus finite-sample statistical bias. We show that the naive plug-in entropy/MI estimator is systematically biased upward at finite event count N , in exactly the direction that could be mistaken for a quantum excess if left uncorrected. Using a controlled Monte Carlo toy model with a known ground truth, we validate two standard corrections (Miller–Madow and shuffle-subtraction) and a more rigorous Bayesian estimator (NSB, Nemenman–Shafee–Bialek), and use the best-behaved of these to derive a concrete, quantitative detection-sensitivity curve: how large a genuine quantum-correlation excess must be, at a given event count, to be statistically distinguishable from finite-sample noise.

Contents

1	Physics motivation	2
2	The observable: MI, conditional entropy, and the quantum witness	2
3	The statistical problem: finite-sample bias	3
4	Toy Monte Carlo test bed	4
4.1	Demonstration: two samples, same true distribution, different size	4
4.2	Bias vs. sample size	5
5	Corrections: Miller–Madow and shuffle-subtraction	5
5.1	Miller–Madow correction	5
5.2	Shuffle-subtraction (label-permutation null)	5
6	The rigorous fix: NSB Bayesian entropy estimator	7
7	Putting it together: quantum-signal detection sensitivity	10
8	Summary and recommendations from the toy Monte Carlo study	12

9	Validation on PYTHIA 8 Monte Carlo simulation	13
9.1	Multiplicity and p_T distributions	13
9.2	The multiplicity- p_T correlation	14
9.3	Finite-sample bias, reproduced on generator output	15
9.4	Point estimates on the full sample	16
9.5	Differential measurement vs. jet p_T	16
9.6	Multi-method comparison vs. p_T	16
10	Overall conclusions and next steps	17

1 Physics motivation

Take a back-to-back dijet event (e.g. from e^+e^- , DIS, or pp collisions at RHIC/STAR). Treat jet A and jet B as the two subsystems of a bipartite system whose “state” is accessed only through observed hadron multiplicities, via **local parton-hadron duality (LPHD)**: the hadron multiplicity distribution stands in for the underlying parton Fock-state distribution.

The Schmidt-decomposition ansatz (Kharzeev-Levin; Peschanski-Seki): if each Fock/parton-number sector n carries probability $P(n)$, and the reduced density matrix of a jet is diagonal in n , then the entanglement entropy of that jet reduces to the **Shannon entropy of its hadron multiplicity distribution**,

$$S_A = - \sum_n P_A(n) \ln P_A(n), \quad (1)$$

and likewise for S_B . This turns a genuinely quantum quantity (entanglement entropy) into something directly measurable from ordinary multiplicity counting – the whole program rests on this ansatz.

The physics question: by additionally measuring the *joint* multiplicity distribution $P_{AB}(n_A, n_B)$ across both jets in the same event, we can construct the mutual information $I(A:B)$ between the two jets (defined in Sec. 2). A large enough MI, in the right combination with the individual entropies, would signal correlations between the jets that cannot be explained by any classical joint probability distribution – i.e. a genuine quantum (entanglement) signature surviving from the partonic state through hadronization, rather than an ordinary classical correlation (e.g. shared color flow, momentum conservation, or other QCD-driven but fundamentally classical correlations).

Why this is subtle in practice: any such claim is only as good as (a) correctly separating genuine quantum correlation from unavoidable classical correlation, and (b) correctly accounting for the finite size of real data samples, since MI estimated from a finite event sample is **always statistically biased**, and – as we show below – biased in exactly the direction that could be mistaken for a quantum excess if not handled carefully. This note works through both issues end to end.

2 The observable: MI, conditional entropy, and the quantum witness

From the per-jet and joint hadron-multiplicity distributions we can build three entropies and two derived quantities:

$$S_A, \quad S_B, \quad S_{AB} \quad \Longrightarrow \quad I(A:B) = S_A + S_B - S_{AB}, \quad S(B|A) = S_{AB} - S_A. \quad (2)$$

Mutual information alone cannot distinguish quantum from classical. $I(A:B) > 0$ just means the jets are correlated *somehow* – a purely classical joint distribution (shared color flow, energy–momentum conservation, any ordinary QCD dynamics) also gives nonzero MI. A sharper diagnostic is needed.

The sharper test: conditional entropy negativity. For *any* classical discrete probability distribution, Shannon conditional entropy obeys

$$S(B|A) = S_{AB} - S_A \geq 0 \quad (\text{always, classically}). \quad (3)$$

This is the classical subadditivity of Shannon entropy – not an assumption about any particular physical model. Only the *quantum* (von Neumann) conditional entropy can go negative, and only for entangled states. So $S(B|A) < 0$ is a **hard classical impossibility**, and observing it (at sufficient statistical significance) would be direct evidence that the jets cannot be described by a classical joint probability distribution – i.e. a genuine quantum signature. Equivalently, in terms of MI alone,

$$I(A:B) > \min(S_A, S_B) \iff S(B|A) < 0 \text{ or } S(A|B) < 0, \quad (4)$$

so MI exceeding the smaller of the two marginal entropies is the same statement in a different form. Both directions ($S(B|A)$ and $S(A|B)$) are worth checking independently, since conditional entropy is not symmetric.

What this test cannot do: genuine quantum discord. True (Ollivier–Zurek) quantum discord is defined by optimizing conditional entropy over *quantum measurement bases* – non-commuting projective measurements that don’t reduce to simply reading off a classical value. Hadron-multiplicity data (N_A, N_B) is, by construction, a classical joint distribution (a histogram). For any classical joint distribution, the optimal “measurement” on B is trivially just observing N_B exactly, which makes the classical-correlation piece $J(A:B)$ equal to the *full* $I(A:B)$ exactly – so discord $D = I - J \equiv 0$ identically, for any classical distribution, with no exception. Extracting a genuinely nonzero discord requires actual quantum-state tomography (e.g. the $\Lambda\bar{\Lambda}$ spin-density-matrix channel), not multiplicity counting. **The conditional-entropy-negativity test above is therefore the correct and complete witness available from this kind of data** – the rest of this note is about measuring it reliably.

3 The statistical problem: finite-sample bias

Every quantity above is estimated from a finite event sample, and the naive (“plug-in”) entropy estimator has a **systematic, structural bias** that turns out to point in exactly the dangerous direction for this measurement.

Where the bias comes from. The naive estimator replaces true probabilities p_i with empirical frequencies $\hat{p}_i = n_i/N$,

$$\hat{S} = - \sum_i \hat{p}_i \ln \hat{p}_i. \quad (5)$$

Because $x \ln x$ is convex (equivalently $-x \ln x$ is concave), Jensen’s inequality guarantees

$$\langle \hat{S} \rangle < S_{\text{true}} \quad \text{for any finite } N, \quad (6)$$

i.e. the plug-in estimator always underestimates entropy. This isn't a numerical artifact; with only N draws one can never fully resolve a distribution's tails, and that missing resolution always looks like *less* apparent randomness than there truly is.

Leading-order size of the effect (Miller 1955; Madow), for a distribution over K possible outcomes:

$$\langle \hat{S} \rangle \approx S_{\text{true}} - \frac{K-1}{2N}. \quad (7)$$

Fewer events (N small) or finer binning (K large) both make the downward bias worse.

Why this flips sign and biases $I(A:B)$ upward. Combining the individual biases of $\hat{S}_A, \hat{S}_B, \hat{S}_{AB}$ with correct signs,

$$\langle \hat{I} \rangle \approx I_{\text{true}} + \frac{(K_{AB}-1) - (K_A-1) - (K_B-1)}{2N} = I_{\text{true}} + \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (8)$$

The joint distribution lives in a much larger space than either marginal alone, so $\hat{S}_{AB}$ suffers a *much larger* downward bias than $\hat{S}_A + \hat{S}_B$ does – and since S_{AB} enters with a **minus** sign, its extra negative bias flips into a **positive** bias on $\hat{I}$. In the extreme case $N \ll K_A K_B$ (badly undersampled joint histogram), this produces large *phantom* apparent mutual information even for truly independent variables – sampling noise alone can masquerade as a real correlation, and by the same mechanism can masquerade as evidence for the “quantum” side of the $S(B|A) < 0$ test above if not corrected.

A commonly-quoted shortcut version of Eq. (8) is $(K_A-1)(K_B-1)/(2N)$; this is not a different formula, it is Eq. (8) evaluated under the extra assumption $K_{AB} = K_A K_B$, i.e. a fully and uniformly occupied joint space. That assumption only holds for independent variables; for correlated data the observed K_{AB} is much smaller than $K_A K_B$ (the joint distribution concentrates on a narrow ridge rather than filling the full grid), so this shortcut badly *overestimates* the bias – a pitfall we hit directly in Sec. 5.

Practical consequence. Comparing the same underlying physics across data samples of different size – different STAR running periods, different luminosity, different jet-cone or multiplicity binning choices – will report systematically different central MI (and $S(B|A)$) values purely from this effect, with zero change in the actual QCD physics. Secs. 4–6 demonstrate this concretely with a toy model and work through three ways to correct for it; Sec. 7 turns the corrected, best-behaved estimator into a quantitative statement about how large a genuine quantum signal would need to be, at a given event count, to be distinguishable from this bias.

4 Toy Monte Carlo test bed

To demonstrate the bias concretely and validate corrections against a known ground truth, we use a controlled toy model: two back-to-back jets, A and B , with a shared latent hard-process fluctuation Λ (e.g. fluctuating parton virtuality / initial-state energy scale) drawn from a Gamma distribution. Conditional on Λ , each jet's hadron multiplicity is Poisson distributed,

$$\Lambda \sim \text{Gamma}(k, \theta), \quad N_A | \Lambda \sim \text{Poisson}(\Lambda), \quad N_B | \Lambda \sim \text{Poisson}(c\Lambda). \quad (9)$$

Marginalizing over Λ gives each jet a **Negative Binomial** multiplicity distribution (the standard KNO-like form used for hadron multiplicities), and the shared Λ induces a genuine – but purely

classical – correlation between N_A and N_B , i.e. $I_{\text{true}} > 0$ with $S(B|A)_{\text{true}}$ solidly positive, exactly as required classically. This is a convenient sandbox: we know the true joint distribution (to arbitrary precision, from a huge reference sample), so we can measure the *bias* of any estimator directly, and later measure how large a genuine quantum excess would need to be to be distinguishable from this classical baseline’s own statistical noise. Parameters used throughout: $k = 3.0$, $\theta = 8.0$, $c = 0.9$ (giving $\text{corr}(N_A, N_B) \approx 0.88$).

4.1 Demonstration: two samples, same true distribution, different size

We generate a “small” event sample ($N = 500$; e.g. a single limited run / low-luminosity STAR-like dataset), a “large” event sample ($N = 200,000$; high-statistics dataset), and a very large reference sample ($N = 5 \times 10^6$) to pin down I_{true} – all drawn from the *exact same underlying physics*.

Table 1: Naive $\hat{I}$ for three samples of the same true physics.

Sample	N	K_A	K_B	K_{AB}	$\hat{I}_{\text{naive}}$ (nats)
small	500	67	60	390	1.7424
large	200,000	129	113	3949	0.7463
reference	5,000,000	159	147	7022	0.7329 ($\approx I_{\text{true}}$)

We see $\hat{I}_{\text{small}} > \hat{I}_{\text{large}} \approx \hat{I}_{\text{ref}} \equiv I_{\text{true}}$ – **the same physics, analyzed with fewer events, reports a systematically larger MI**. This is the bias predicted in Sec. 3, not a change in physics.

The bias comes entirely from how well-populated the joint histogram is. Figure 1 shows (top) the marginal hadron-multiplicity distributions $P_A(n)$, overlaid for the three samples – these look reasonably similar even at $N = 500$ because there are only $O(10^2)$ bins to fill – and (bottom) the 2D joint histograms $P_{AB}(n_A, n_B)$ on the same color scale, where the difference is dramatic: at $N = 500$ the joint plane is almost empty, while at $N = 5 \times 10^6$ it is a smooth, fully-resolved correlated ridge. A sparse, “salt-and-pepper” joint histogram is exactly what drives $\hat{S}_{AB}$ artificially low, and hence $\hat{I}$ artificially high.

4.2 Bias vs. sample size

Repeating the naive estimate over a range of N (with several repetitions per N to see the scatter) and plotting $\hat{I}_{\text{naive}}$ against $1/N$ (Fig. 2) shows the systematic downward drift (bias shrinking) as N grows, converging toward the flat reference line, and the expected near-linear behavior in $1/N$: the intercept of that line is a cheap, model-independent estimate of the bias-corrected/true MI. The extrapolated intercept, 0.9427 nats, lands reasonably close to the reference value of 0.7329 nats.

5 Corrections: Miller–Madow and shuffle-subtraction

5.1 Miller–Madow correction

The exact leading-order Miller–Madow correction applies the $-(K - 1)/(2N)$ term to *each* of S_A, S_B, S_{AB} separately and combines them:

$$\hat{I}_{MM} = \hat{I}_{\text{naive}} - \frac{(K_{AB} - 1) - (K_A - 1) - (K_B - 1)}{2N} = \hat{I}_{\text{naive}} - \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (10)$$

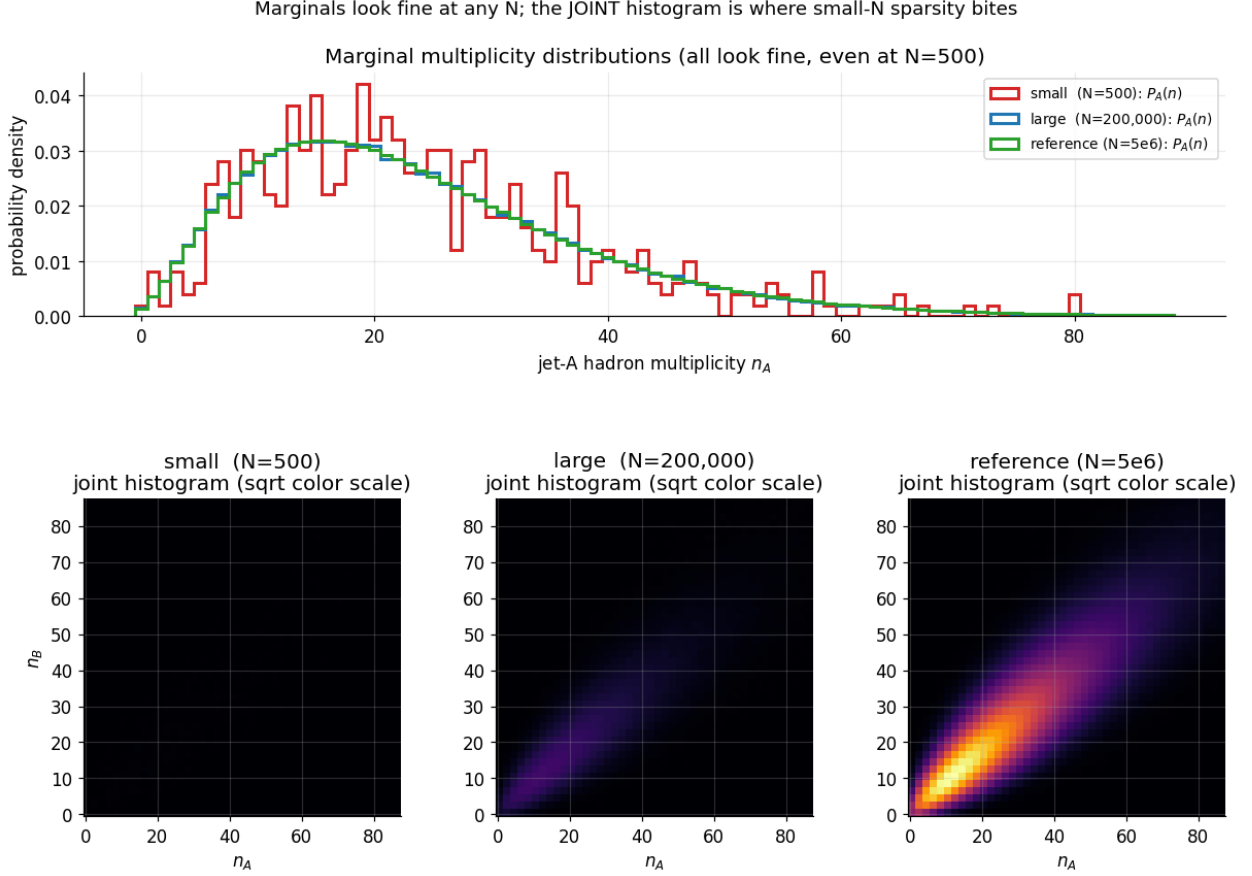


Figure 1: Marginal multiplicity distributions (top) and 2D joint histograms (bottom, square-root color scale) for the small, large, and reference samples. Marginals look consistent at any N ; the joint histogram is where small- N sparsity bites.

Crucially, K_{AB} here must be the *actually observed* number of occupied joint bins, not the product $K_A K_B$: for correlated variables (our case, by construction) the joint distribution concentrates on a narrow ridge and $K_{AB} \ll K_A K_B$, so using the product in place of the observed value wildly overcorrects and can drive $\hat{I}_{MM}$ negative.

5.2 Shuffle-subtraction (label-permutation null)

Randomly permuting the B -jet labels relative to the A -jet labels *within the same sample* destroys any true correlation ($I_{\text{true, shuffled}} = 0$ by construction) while reproducing the same finite-sample bias. Averaging over many shuffles gives a direct empirical estimate of the bias to subtract:

$$\hat{I}_{\text{corrected}} = \hat{I}_{\text{naive}} - \langle \hat{I}_{\text{shuffled}} \rangle. \quad (11)$$

This makes no asymptotic assumption and remains reliable even when $N \lesssim K_A K_B$, where Miller–Madow starts to break down. It has its own bias, however: shuffling spreads the joint distribution over *more* bins than the true correlated data occupies, so the shuffled surrogate is *more* under-sampled than the real data at the same N . Shuffle-subtraction therefore tends to systematically **overestimate** the bias and **overcorrect**, pulling $\hat{I}_{\text{corrected}}$ below I_{true} when the true correlation is strong.

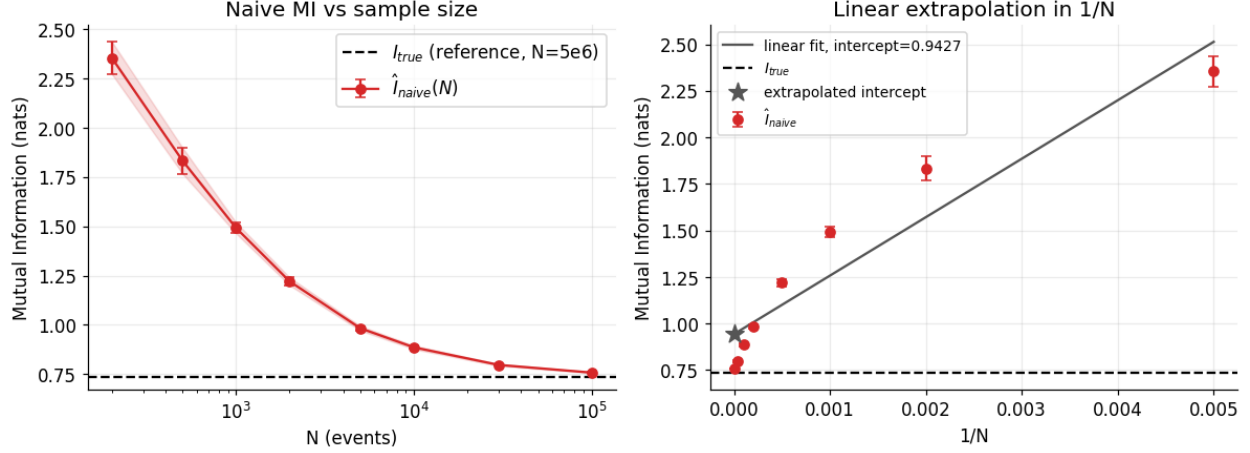


Figure 2: Left: naive $\hat{I}$ vs. sample size N , converging to the reference value. Right: the same data plotted against $1/N$, with a linear extrapolation to $1/N \rightarrow 0$.

Table 2: Naive, Miller–Madow, and shuffle-corrected $\hat{I}$ vs. N .

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	I_{true} (ref.)
200	2.2764	2.0889	0.0657	0.7329
500	1.8694	1.6014	0.1436	0.7329
1000	1.4271	1.1756	0.2422	0.7329
2000	1.1873	0.9866	0.3809	0.7329
5000	0.9766	0.8539	0.5592	0.7329
20000	0.8259	0.7750	0.6792	0.7329

With the correct Miller–Madow formula (using the observed K_{AB}), both corrections pull the small- N estimates much closer to I_{true} than the naive value and stay well-behaved (no more diving negative). Neither lands exactly on I_{true} at very small N : Miller–Madow is only a first-order truncation of the true bias expansion, and shuffle-subtraction overcorrects when the true correlation is strong, as explained above. Both effects shrink as N grows.

The shuffle procedure gives more than a bias correction – the spread of $\hat{I}_{\text{shuffle}}$ over many permutations is the null distribution for “ $I_{\text{true}} = 0$ ” at that N and binning (Fig. 3). Plotting $\hat{I}_{\text{naive}}$ against that null distribution is a direct visual significance check: at $N = 500$ the null distribution itself is already centered well above zero (pure bias) and the naive value sits close to or inside this null cloud, so on its own it would not be significant evidence of correlation, even though the jets are truly correlated by construction. By $N = 20,000$ the null has collapsed toward zero and the naive value clearly separates from it.

Figure 4 summarizes naive vs. corrected MI across sample sizes, and Fig. 5 plots $|\hat{I} - I_{\text{true}}|$ on a log-log scale, which is the honest way to compare convergence rates. Both corrections sit visibly below the naive curve at fixed N ; where the corrected residual curves run roughly parallel to the naive curve rather than dropping faster, that indicates the residual error is increasingly dominated by sub-leading bias terms that neither method captures (Miller–Madow: higher-order $1/N^2, \dots$ terms; shuffle: the systematic overcorrection discussed above) – exactly the gap the more rigorous fix in Sec. 6 is designed to close.

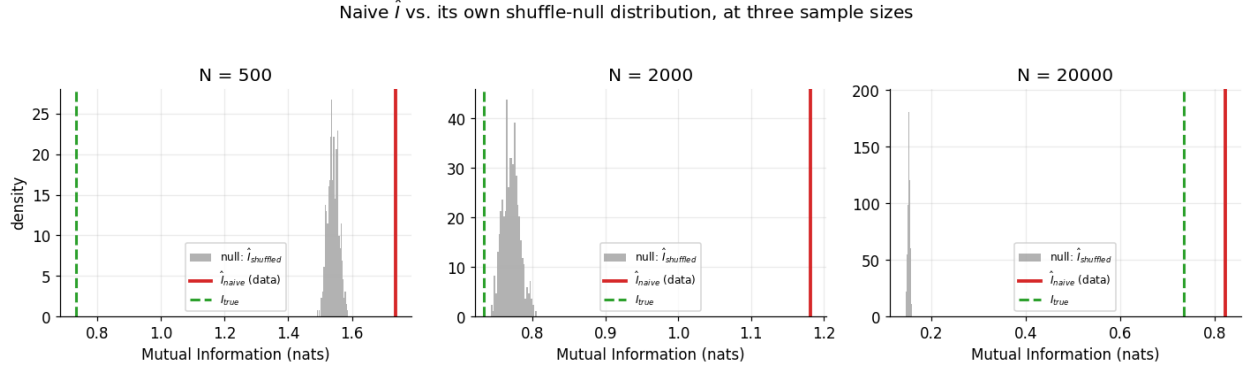


Figure 3: Naive $\hat{I}$ (vertical line) against its own shuffle-null distribution (histogram), at three sample sizes. Dashed line: true value.

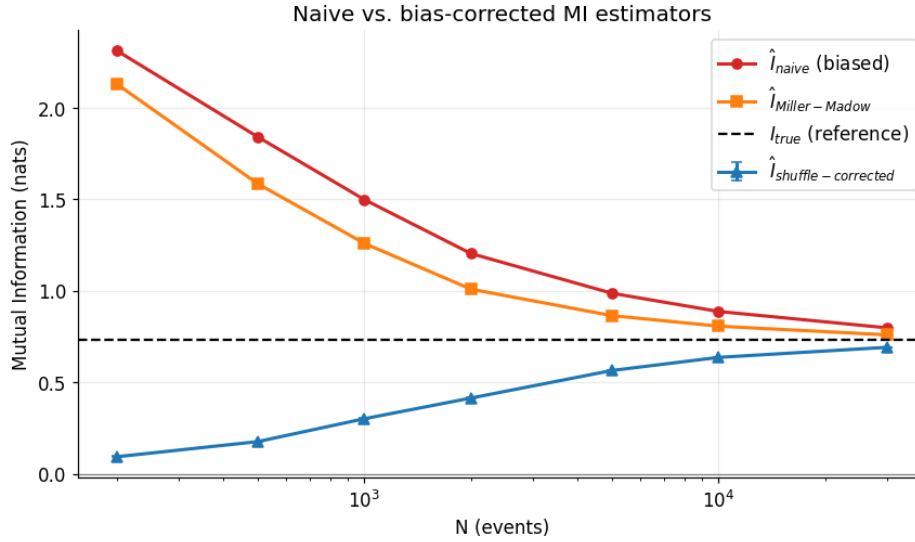


Figure 4: Naive vs. bias-corrected MI estimators across sample sizes.

6 The rigorous fix: NSB Bayesian entropy estimator

Both corrections above are still, fundamentally, **point corrections derived under a large- N assumption** – they estimate one number (the bias) and subtract it. The **Nemenman–Shafee–Bialek (NSB)** estimator takes a genuinely different approach: instead of correcting a biased point estimate, it computes the **Bayesian posterior mean entropy** directly, using a prior specifically constructed to avoid smuggling in any implicit assumption about how spread-out the distribution is. This is the standard tool for the severely-undersampled regime ($N \lesssim K$) in the neuroscience spike-train MI literature – precisely our regime once hadron multiplicities are binned in 2D.

The construction, in four steps:

1. Put a symmetric Dirichlet prior $P(p|\beta) \propto \prod_i p_i^{\beta-1}$ over the K possible bin probabilities. For any fixed β , the posterior mean entropy given counts $\{n_i\}$ has a closed form,

$$\langle S \rangle_{\beta,n} = \psi_0(N + K\beta + 1) - \sum_i \frac{n_i + \beta}{N + K\beta} \psi_0(n_i + \beta + 1). \quad (12)$$

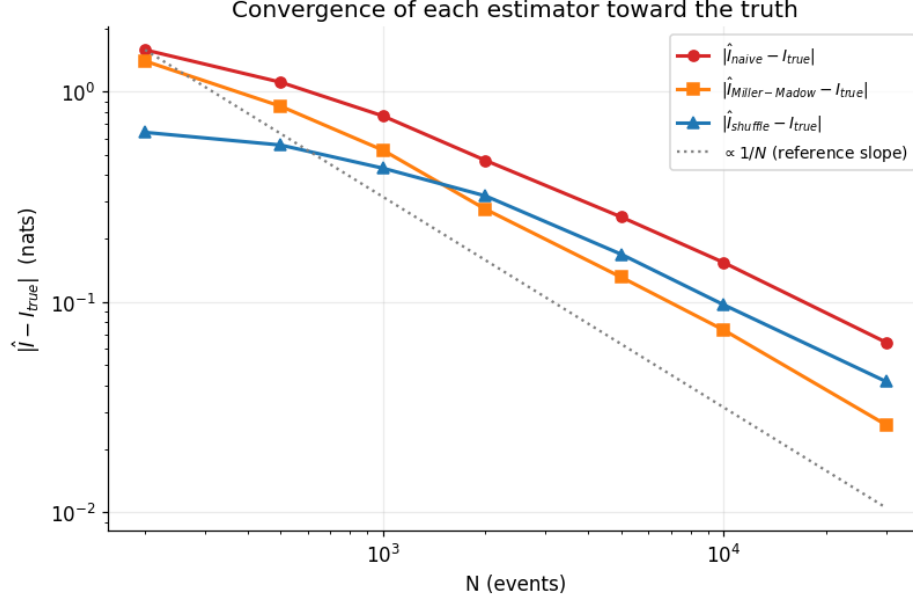


Figure 5: Convergence of each estimator toward the true value, on a log-log scale, with a $1/N$ reference slope.

2. **The catch:** a flat prior on β does *not* give a flat prior on entropy – for large K , $P(S|\beta)$ is sharply peaked as a function of β . Fixing any single β (as ordinary add-one/Laplace/Jeffreys smoothing does) silently assumes an entropy value before looking at the data.
3. NSB’s fix: reweight β by the hyperprior that makes the *induced* prior on S flat – literally the rate of change of the prior mean entropy with β ,

$$\rho(\beta) = K \psi_1(K\beta + 1) - \psi_1(\beta + 1). \quad (13)$$

4. Average over β , weighted by both this hyperprior and the Dirichlet–multinomial evidence $P(n|\beta)$,

$$\hat{S}_{\text{NSB}} = \frac{\int_0^\infty d\beta \rho(\beta) P(n|\beta) \langle S \rangle_{\beta,n}}{\int_0^\infty d\beta \rho(\beta) P(n|\beta)}. \quad (14)$$

This reduces to a tractable 1D numerical integral. We apply it separately to S_A, S_B, S_{AB} (each needs its own alphabet size K : for the joint distribution, $K_{AB}^{\text{alphabet}} = K_A^{\text{alphabet}} \times K_B^{\text{alphabet}}$, the full product space of possible outcomes – not the observed K_{AB} count from before, since NSB needs to know the size of the space being integrated over, not how much of it happened to be hit), then combine: $\hat{I}_{\text{NSB}} = \hat{S}_A^{\text{NSB}} + \hat{S}_B^{\text{NSB}} - \hat{S}_{AB}^{\text{NSB}}$.

NSB tracks I_{true} closely across the entire range, including at $N = 200$ where the naive estimator is off by roughly a factor of three and Miller–Madow has barely moved. This is the payoff of treating the whole problem as Bayesian inference over entropy rather than patching a biased point estimate after the fact. Figure 6 adds NSB to the convergence comparison: its residual curve sits visibly below the other three methods across most of the range, and in particular stays far more stable at small N – it doesn’t need $N \gg K$ to behave sensibly.

Table 3: All four MI estimators, head to head.

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	$\hat{I}_{\text{NSB}}$	I_{true} (ref.)
200	2.3511	2.1711	0.0578	0.5098	0.7329
500	1.7525	1.5035	0.2137	0.6833	0.7329
1000	1.5071	1.2711	0.3062	0.7369	0.7329
2000	1.2299	1.0367	0.4340	0.7416	0.7329
5000	1.0030	0.8842	0.5807	0.7678	0.7329
20000	0.8222	0.7716	0.6752	0.7411	0.7329

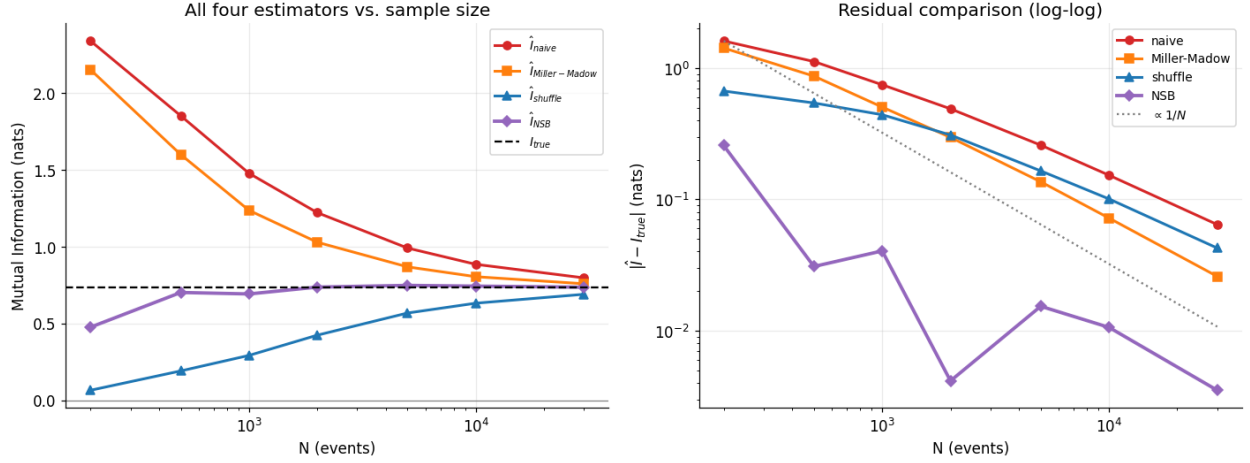


Figure 6: Left: all four estimators vs. sample size. Right: residual comparison on a log-log scale, with a $1/N$ reference slope.

Practical notes for the real STAR dijet data.

- The alphabet size K (for each marginal, and $K_A \times K_B$ for the joint) needs to be fixed *before* looking at the data – pick a generous fixed upper bound on plausible hadron multiplicity per jet (e.g. from detector acceptance / kinematic limits), not the observed range, or a data-dependent bias is reintroduced through the back door.
- NSB is somewhat more expensive per call than MM or shuffle-subtraction (a 1D numerical integral instead of a closed-form correction or a handful of permutations), but it remains a fraction of a second per event sample – entirely practical for a real analysis, including a full jackknife or bootstrap for uncertainty quantification.
- The same estimator applies directly to $S(B|A) = S_{AB} - S_A$ – swap in the NSB entropies there instead of the naive/MM/shuffle versions before making any negative-conditional-entropy (quantum) claim, as done next.

7 Putting it together: quantum-signal detection sensitivity

Section 2 established $S(B|A) = S_{AB} - S_A$ as the correct, computable witness (provably ≥ 0 classically), and Secs. 3–6 established that NSB is the best-behaved bias-corrected estimator, especially at small N . Here we combine both: since our toy model is itself purely classical, its true $S(B|A)$

is a known, solidly positive number ($S(B|A)_{\text{true}} = 3.1524$ nats) – exactly the right sandbox to measure the **statistical noise floor** of the NSB estimator against a known ground truth, and turn that into a quantitative statement about how large a genuine quantum excess would need to be, at a given event count, to be distinguishable from this noise floor.

Design: for a grid of N , we draw many independent samples from the same generative model, compute $\hat{S}(B|A)_{\text{NSB}}$ for each, and take the spread over draws as the estimator’s sampling uncertainty $\sigma_{\text{NSB}}(N)$ at that statistics. A hypothetical genuine quantum excess of magnitude Δ is then detectable at z standard deviations once

$$\Delta > z \sigma_{\text{NSB}}(N). \quad (15)$$

Equivalently, the **minimum detectable quantum signal** at a chosen confidence level ($z = 3$, a common “solid evidence” threshold, used below) is $\Delta_{\text{min}}(N) = z \sigma_{\text{NSB}}(N)$. In a real analysis with one actual dataset – not a synthetic model with known truth – $\sigma_{\text{NSB}}(N)$ would be estimated directly from that dataset via bootstrap or jackknife resampling, rather than from independent Monte Carlo replicates as done here for demonstration.

Figure 7 shows the NSB (and, for comparison, naive and Miller–Madow) estimates of $S(B|A)$ vs. N , together with their sampling standard deviation. NSB has both the smallest bias (its mean tracks $S(B|A)_{\text{true}}$ most closely, especially at small N) and a noise floor that shrinks cleanly with N , following the expected $1/\sqrt{N}$ statistical scaling.

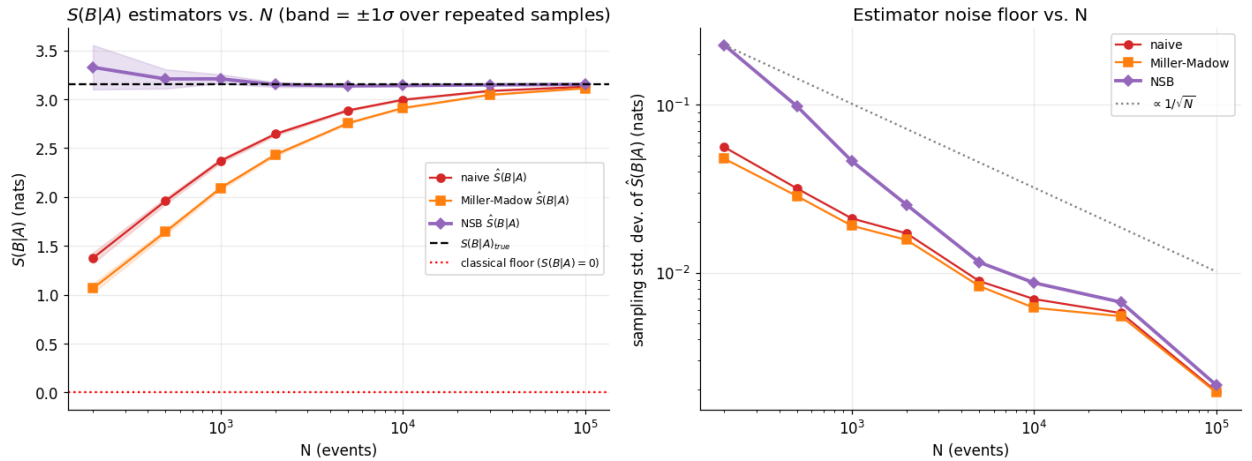


Figure 7: Left: $S(B|A)$ estimators vs. N (bands: $\pm 1\sigma$ over repeated samples). Right: sampling standard deviation of each estimator vs. N , with a $1/\sqrt{N}$ reference slope.

Converting the NSB noise floor into a minimum-detectable-signal curve (Fig. 8) and fitting a power law gives a direct, quantitative answer to “how big must a quantum signal be, at a given statistics, to be believable”:

Caveats to carry into the real analysis.

- $\sigma_{\text{NSB}}(N)$ here is measured *at the true classical value* ($S(B|A)_{\text{true}} \approx$ a few nats for our toy correlation strength), not at the boundary $S(B|A) = 0$. Entropy-estimator variance depends on the shape of the underlying distribution, not just N , so the true noise floor near a genuinely small or slightly-negative $S(B|A)$ could differ somewhat from this estimate. For the real STAR data, this bootstrap procedure should be re-run centered on the actual observed operating

Table 4: Minimum event count N needed for a 3σ detection of a hypothetical quantum-correlation excess of the given magnitude, from the fitted $\Delta_{\min}(N) \propto N^{-0.72}$ power law.

Assumed $ S(B A)_{\text{quantum}} $ (nats)	N needed for 3σ detection
0.500	201
0.200	722
0.100	1,899
0.050	4,995
0.020	17,934
0.010	47,164

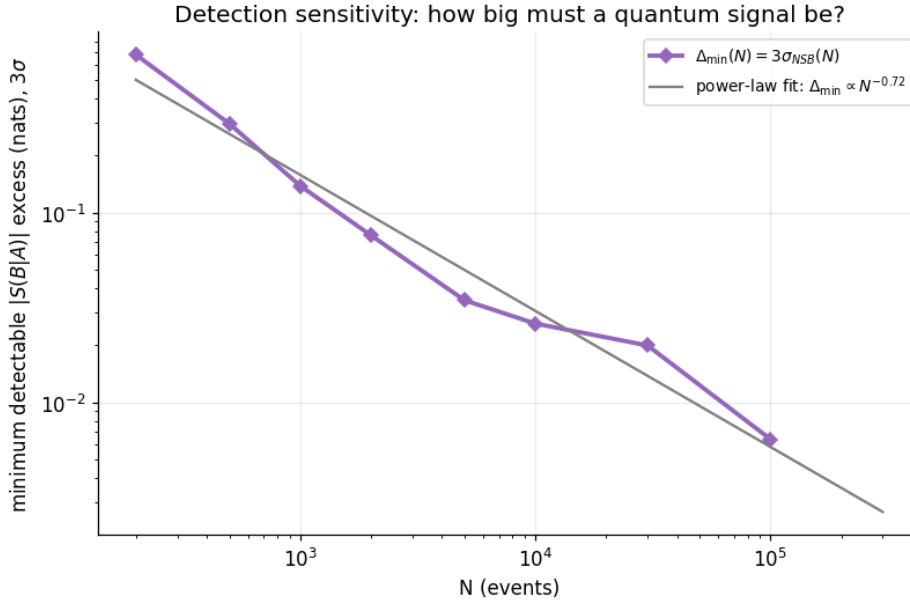


Figure 8: Minimum detectable $|S(B|A)|$ excess at 3σ , vs. event count N , with a power-law fit for interpolation/extrapolation.

point (via jackknife/bootstrap on the real dataset itself), rather than trusting this toy-model number directly.

- This procedure characterizes sensitivity to a violation of the classical conditional-entropy bound using multiplicity data alone (Sec. 2). It is not a substitute for an actual quantum witness such as a Bell/CHSH inequality on the $\Lambda\bar{\Lambda}$ spin observables, which remains the model-independent way to certify entanglement rather than merely rule out one particular classical description.

8 Summary and recommendations from the toy Monte Carlo study

1. **The correct witness** is $S(B|A) = S_{AB} - S_A$ (and $S(A|B)$, check both directions), which must be ≥ 0 for any classical distribution. Genuine quantum discord (Ollivier–Zurek) cannot be computed from hadron-count histograms alone – any classical joint distribution trivially gives discord = 0; it would require actual quantum-state tomography (e.g. the $\Lambda\bar{\Lambda}$ spin-

density-matrix channel).

2. **The naive plug-in MI estimator is always biased upward**, with leading term $\approx (K_{AB} - K_A - K_B + 1)/(2N)$. Coarser binning or fewer events both make it worse – comparing different STAR running periods, luminosities, or binning choices will report different central MI values from what should be the same physics, purely from this effect.
3. **Miller–Madow** (done correctly, using the observed K_{AB} , not the product $K_A K_B$) and **shuffle-subtraction** both substantially reduce this bias and stay well-behaved, but neither is exact at small N : Miller–Madow is a first-order truncation; shuffle-subtraction tends to systematically overcorrect when the true correlation is strong.
4. **NSB** is the recommended estimator for the real analysis: a Bayesian posterior-mean entropy built on a prior specifically constructed to be uninformative about entropy itself. It tracks the truth closely across the entire tested N range, including deep in the severely-undersampled regime where the other methods struggle, at a modest computational cost.
5. **Before claiming any quantum-correlation excess**, bias-correct with NSB and compare against the estimator’s own sampling noise floor at that N (bootstrapped from the real dataset via jackknife/bootstrap, not assumed from this toy model). Section 7 works this out quantitatively: the NSB noise floor shrinks $\propto 1/\sqrt{N}$, giving a concrete minimum-detectable-signal curve $\Delta_{\min}(N) = z \sigma_{\text{NSB}}(N)$ – a direct, quotable answer to “how big must a genuine quantum excess be, at a given event count, to be distinguishable from finite-sample bias and noise.”
6. **This entire procedure only tests a violation of the classical conditional-entropy bound**. It is a necessary check before any stronger quantum claim, but is not itself a model-independent entanglement certification – that role is played by the Bell/CHSH-type inequality on spin observables in the parallel $\Lambda\bar{\Lambda}$ program.

9 Validation on PYTHIA 8 Monte Carlo simulation

The toy model of Secs. 4–6 demonstrates that the estimator pipeline behaves correctly on data with a known, simple generative structure. The natural next step is to check the same pipeline on a full QCD event generator, where hadron multiplicities arise from parton showering, hadronization, and (optionally) the underlying event – none of which was built into the toy model by hand. We generated back-to-back charged-jet dijets with PYTHIA 8 (Monash 2013 tune, anti- k_T jets, $R = 0.4$, charged final-state particles only, matching a STAR-TPC-like charged-jet definition) and applied the identical naive/ Miller–Madow/shuffle/NSB pipeline, unmodified, to the generator output.

This particular validation run used LHC-like kinematics ($\sqrt{s} = 14$ TeV rather than the $\sqrt{s} = 200$ GeV RHIC energy that motivates the eventual STAR measurement) specifically *because* of its much wider accessible jet- p_T range: as discussed below (Sec. 9.2), the multiplicity-vs- p_T correlation predicted by QCD is intrinsically a slow, logarithmic effect, and a wide dynamic range makes it far easier to confirm the pipeline is tracking real physics rather than an artifact of a narrow kinematic window. The 200 GeV run for the actual STAR measurement is a separate, ongoing exercise (see Sec. 10); everything in this section is a *methods validation*, not a physics result.

9.1 Multiplicity and p_T distributions

Figure 9 shows the charged-multiplicity distributions for the leading and subleading jets, overlaid with a method-of-moments Negative-Binomial fit (as in Sec. 4). Both jets show genuine overdispersion relative to Poisson ($\text{var}/\text{mean} = 1.55$ for the leading jet, 1.29 for the subleading jet), qualitatively consistent with – though not identical in shape to – the NBD-like multiplicity distributions assumed in the toy model. This is a useful cross-check in its own right: the toy model’s central structural assumption (NBD-shaped marginals) is not merely a convenient fiction, but a reasonable approximation to what a full QCD generator actually produces.

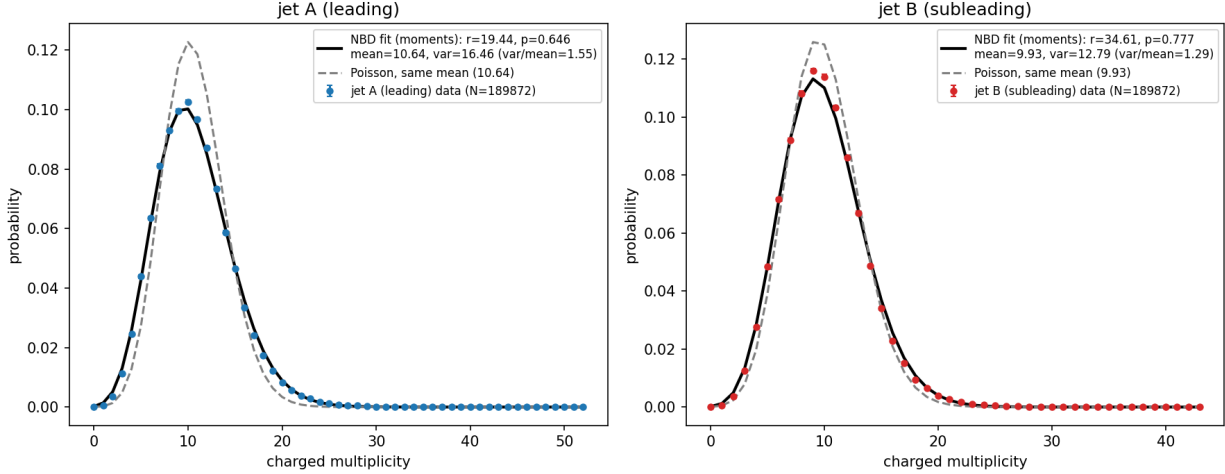


Figure 9: PYTHIA 8: charged-multiplicity distributions for the leading (left) and subleading (right) jet, with NBD and same-mean Poisson overlay (cf. the toy-model version, Sec. 4).

Figure 10 shows the corresponding joint distribution: 36.9% of the 53×44 grid is occupied ($K_{AB} = 861$) at $N \approx 1.9 \times 10^5$ events – a moderately, not severely, undersampled regime, comparable to the middle of the range explored in the toy study. Figure 11 shows the (steeply falling, as expected for a QCD jet spectrum) leading- and subleading-jet p_T distributions, and Fig. 12 their joint distribution – the latter makes visually clear why quantile (rather than fixed-width) p_T binning is necessary for the differential measurements below: the bulk of events sit in a narrow low- p_T band, with a long, sparse tail.

9.2 The multiplicity– p_T correlation

A basic physical expectation – confirmed experimentally (e.g. ALICE, arXiv:2311.13322: “the mean charged-particle multiplicity inside the leading jet cone rises monotonically with increasing jet p_T ”) – is that higher- p_T jets should carry higher charged multiplicity, via additional phase space for parton radiation before hadronization. This growth is predicted by QCD coherent-branching (MLLA) arguments to be slow and roughly *logarithmic* in p_T , not linear, which matters directly for how easy the effect is to see over a limited kinematic range (as it was, initially, in an earlier 200 GeV validation attempt with a narrower accessible p_T window).

Figure 13 shows the raw, unbinned-by-any-entropy- machinery profile of mean charged multiplicity vs. jet p_T , together with a weighted fit of the MLLA-motivated form $\langle N \rangle = a + b \ln p_T$. The fitted slope is highly significant for both jets ($b = 3.77 \pm 0.05$, 83σ , leading jet; $b = 4.71 \pm 0.07$, 66σ , subleading jet) – direct, quantitative confirmation that PYTHIA 8 reproduces the expected

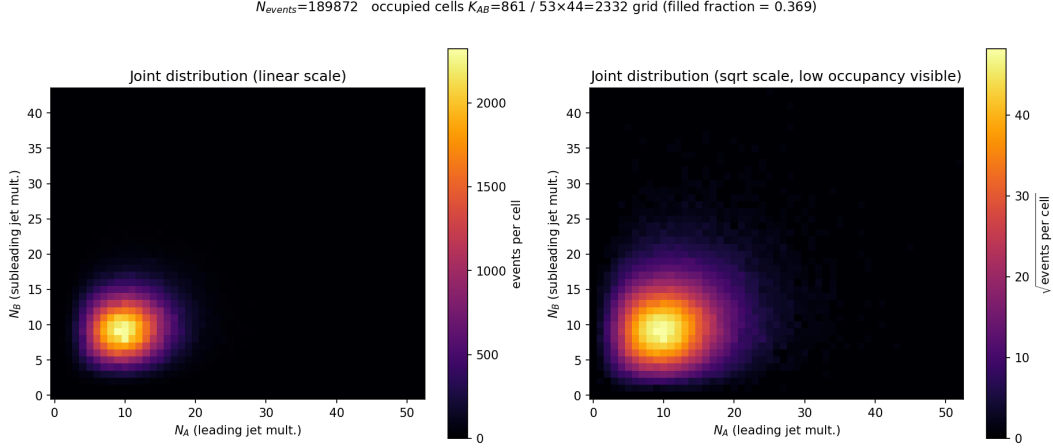


Figure 10: PYTHIA 8: joint charged-multiplicity distribution, linear and sqrt color scale (cf. Fig. 1).

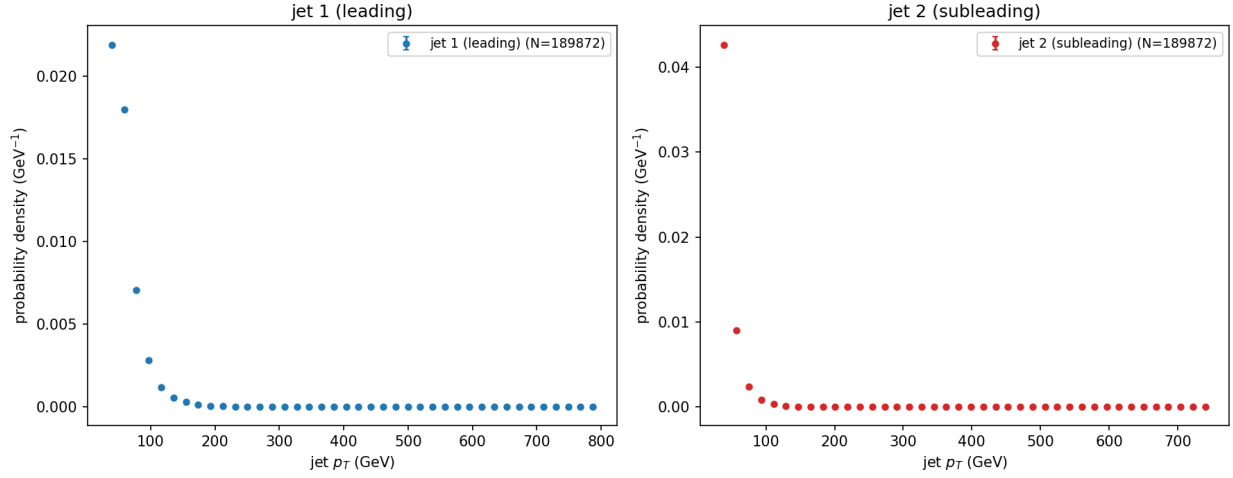


Figure 11: PYTHIA 8: leading- and subleading-jet p_T distributions.

QCD multiplicity-growth pattern, and that the analysis pipeline correctly recovers it from generator output with no prior assumption beyond bin means and standard errors.

9.3 Finite-sample bias, reproduced on generator output

Figure 14 repeats the toy study’s core diagnostic (Sec. 4) directly on subsamples of the real PYTHIA 8 output: the naive $\hat{I}$ falls from ≈ 0.66 nats at $N = 200$ down toward the full-sample NSB value (≈ 0.009 nats) as N grows to $\approx 1.9 \times 10^5$ – the same qualitative bias curve seen throughout Sec. 4, now confirmed on a dataset the toy model was never built to resemble. This is a meaningful cross-check: the bias mechanism identified analytically (Sec. 3) and demonstrated on a synthetic Gamma-Poisson model is not an artifact of that specific generative structure – it appears with the same shape on genuine QCD Monte Carlo output.

N_events = 189872

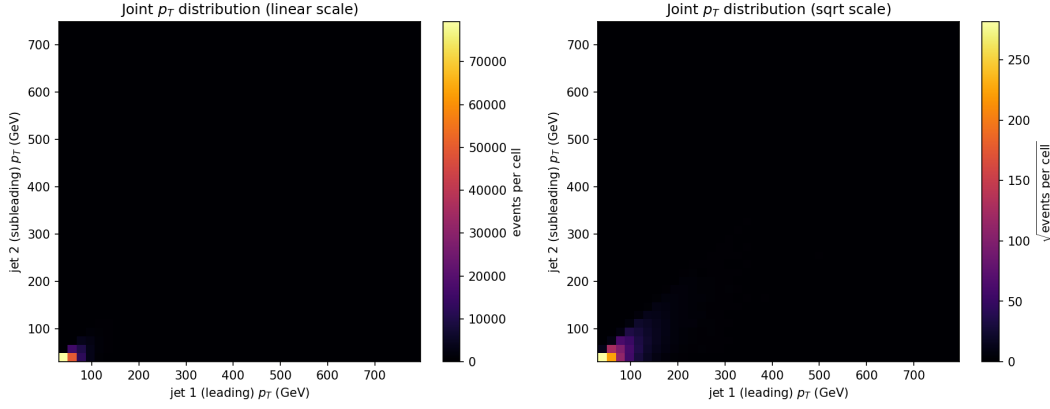


Figure 12: PYTHIA 8: joint leading/subleading jet p_T distribution, linear and sqrt color scale.

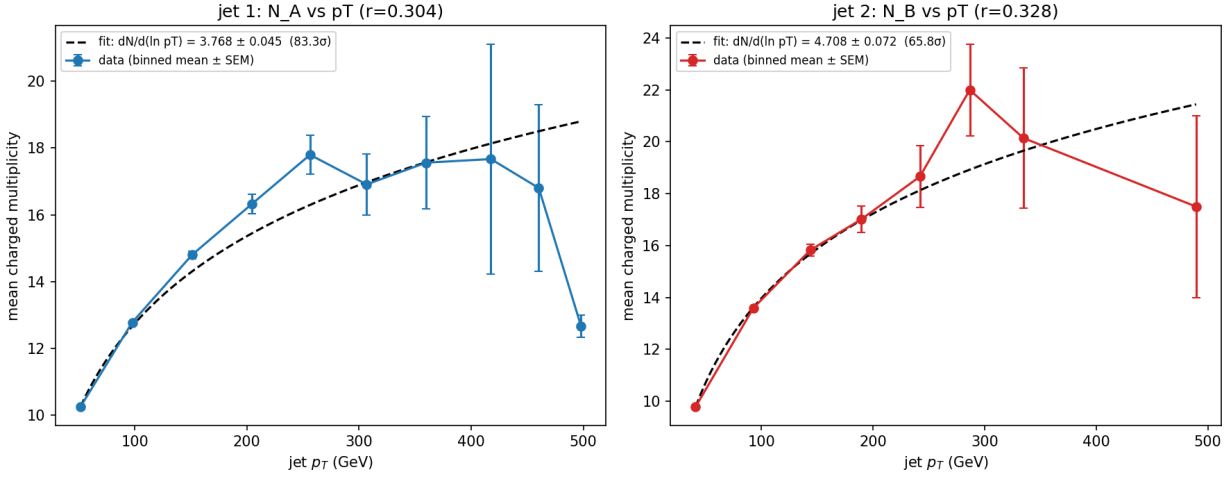


Figure 13: PYTHIA 8: raw mean charged multiplicity vs. jet p_T (binned mean $\pm$ SEM), with a weighted fit to $\langle N \rangle = a + b \ln p_T$. Both slopes are significant at $> 60\sigma$.

9.4 Point estimates on the full sample

Figure 15 shows all four estimators on the full sample ($N \approx 1.9 \times 10^5$): naive (0.0112 ± 0.0004), Miller–Madow (0.0092 ± 0.0004), shuffle-corrected (0.0084 ± 0.0003), and NSB (0.0085 ± 0.0004 nats) – naive sits visibly above the other three, as expected from the bias mechanism above, while Miller–Madow/shuffle/NSB agree closely with each other, consistent with all three having converged toward the true value at this statistics (matching the closure seen among independently-biased estimators at large N in the toy study, Sec. 4). The conditional entropy $S(B|A) = 2.65$ nats sits far above the classical floor, with no indication of a quantum violation – expected and unsurprising, since PYTHIA 8 is itself a classical event-by-event simulator with no entanglement built into its hadronization model. This measurement is a validation of the estimator machinery, not a physics claim; the analogous measurement on real detector data is where a genuine quantum question would be asked.

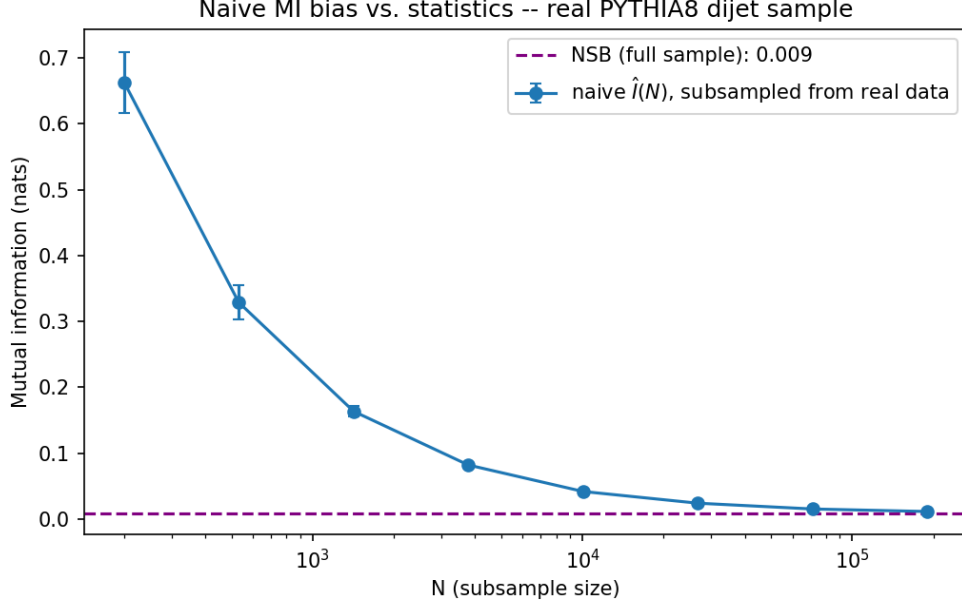


Figure 14: PYTHIA 8: naive $\hat{I}$ vs. subsample size N , real generator output (cf. the toy-model version, Fig. 2).

9.5 Differential measurement vs. jet p_T

Figures 16–18 show S_A, S_B, S_{AB} , $I(A:B)$, and $S(B|A)$ (NSB) in quantile bins of leading-jet p_T . All three entropies rise smoothly with p_T (Fig. 16), directly reflecting the multiplicity growth of Sec. 9.2. $I(A:B)$ (Fig. 17) also rises with p_T , modestly at low-to-mid p_T and more sharply in the highest (widest, most statistics-limited) bin; $S(B|A)$ (Fig. 18) stays solidly positive and rising throughout, consistent with a classical description at every p_T probed.

9.6 Multi-method comparison vs. p_T

Figure 19 overlays naive/Miller–Madow/ shuffle/NSB for $I(A:B)$, and naive/Miller–Madow/NSB for $S(B|A)$ (no shuffle analogue, Sec. 6), each with bootstrap error bars computed by the same unified procedure (Sec. 7). The bias hierarchy naive $>$ Miller–Madow $\gtrsim$ NSB $\approx$ shuffle seen on the full sample (Fig. 15) persists across p_T bins, most visibly in the highest- p_T (lowest-statistics) bin, where naive’s upward bias is largest – exactly the pattern the toy study predicts for a bin with fewer events and a wider effective alphabet.

10 Overall conclusions and next steps

The toy Monte Carlo study (Secs. 4–7) establishes the estimator methodology and its failure modes under controlled, known conditions. Section 9 shows the same pipeline, unmodified, correctly recovers physically-expected behavior (NBD-like overdispersion, MLLA-consistent multiplicity growth with p_T , the same finite-sample bias shape) on realistic QCD Monte-Carlo output where no such structure was hand-built in – a meaningful, if indirect, validation that the pipeline is measuring real structure rather than an artifact of the toy model’s specific construction.

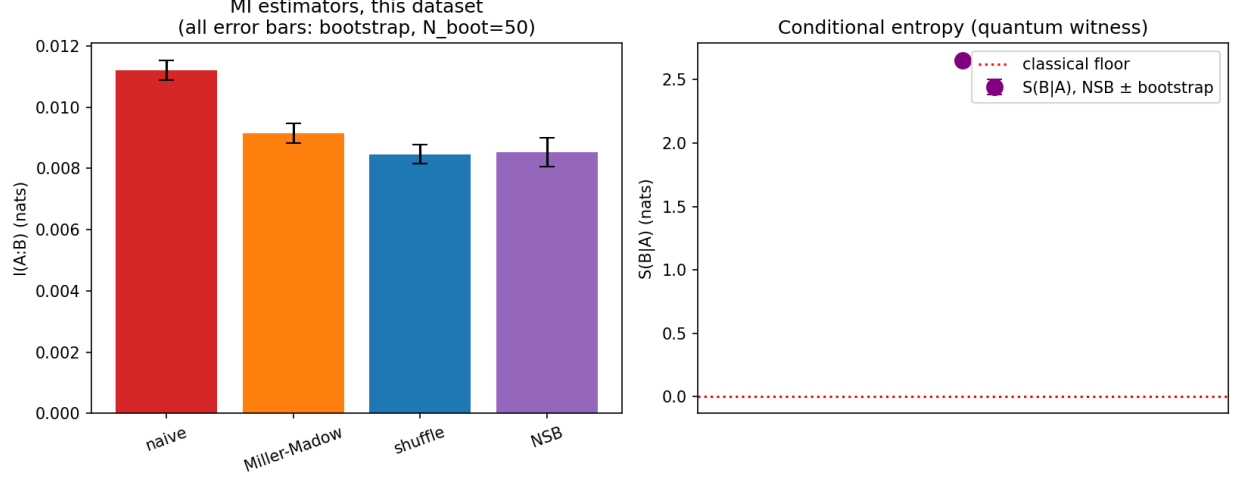


Figure 15: PYTHIA 8: all four MI estimators (left) and $S(B|A)$ (right) on the full sample, bootstrap error bars.

What this note does *not* yet provide is a validation against a dataset with an independently *known* I_{true} that is also calibrated to resemble real jet data (Sec. 9's PYTHIA 8 sample has no known truth of its own). A closure test addressing this gap – fitting the toy model's Gamma-Poisson structure to real PYTHIA 8 marginals and correlation strength, generating a very large reference sample from that fitted model to obtain a calibrated pseudo-truth, and checking recovery (bias and pull-distribution calibration) of all four estimators at the real dataset's actual statistics – is in progress at the time of writing and will be reported separately.

Remaining steps toward the actual STAR measurement: (i) repeat Sec. 9 at $\sqrt{s} = 200$ GeV with STAR-appropriate jet kinematics rather than the wide-dynamic-range 14 TeV validation kinematics used here; (ii) apply the detector-level acceptance and tracking-efficiency model (already implemented in the generator, not yet exercised in this note) and repeat the full analysis chain; (iii) once (i)–(ii) are validated, apply the identical pipeline to real STAR data, with the sensitivity/detection-threshold machinery of Sec. 7 used to state explicitly, before unblinding, how large a genuine quantum-correlation excess would need to be to constitute a significant detection at the actual achieved statistics.

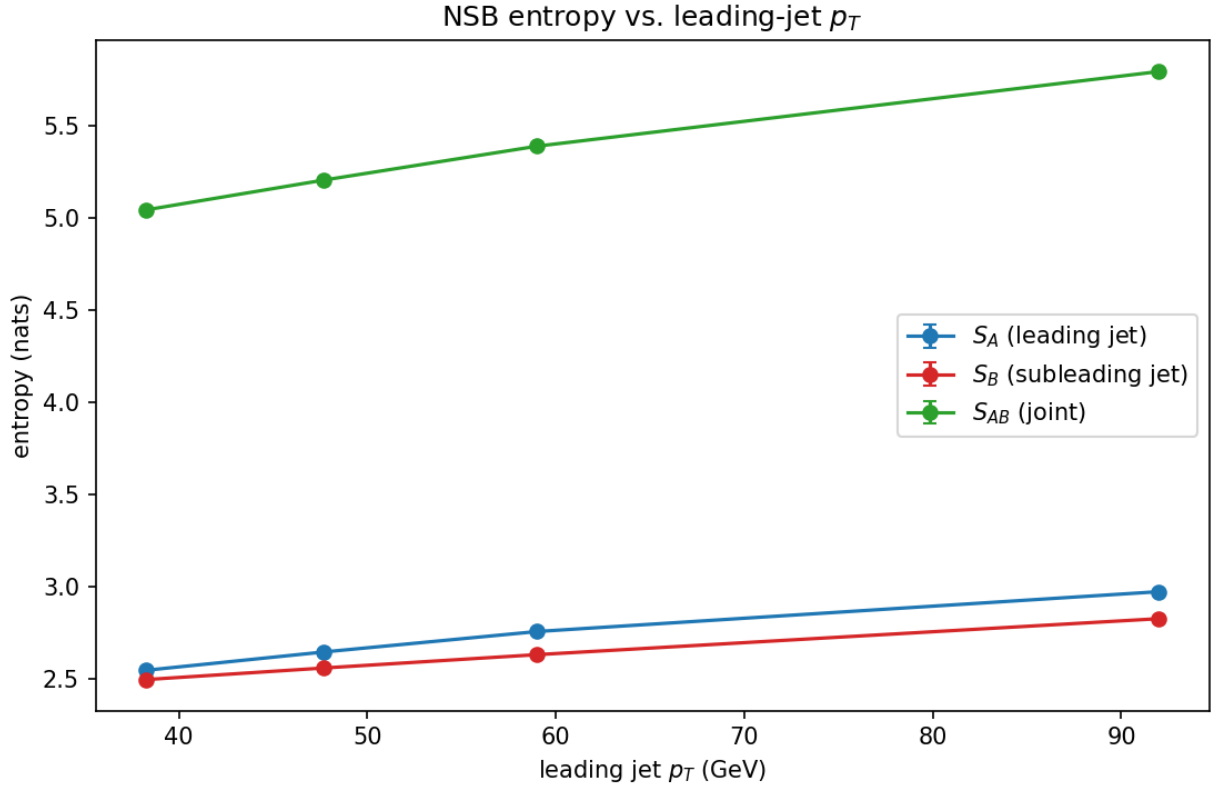


Figure 16: PYTHIA 8: S_A, S_B, S_{AB} (NSB) vs. leading-jet p_T .

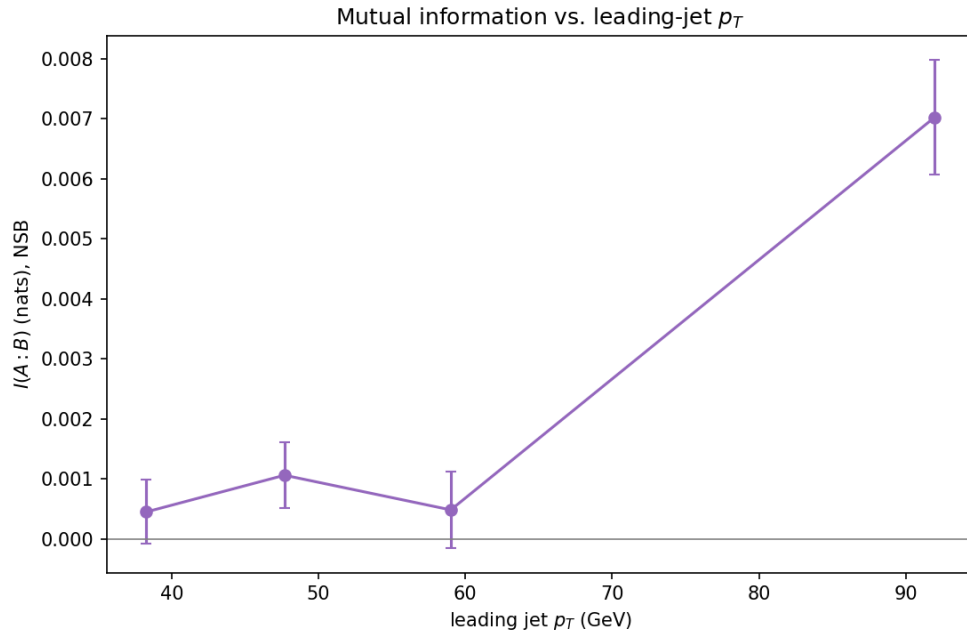


Figure 17: PYTHIA 8: $I(A:B)$ (NSB) vs. leading-jet p_T .

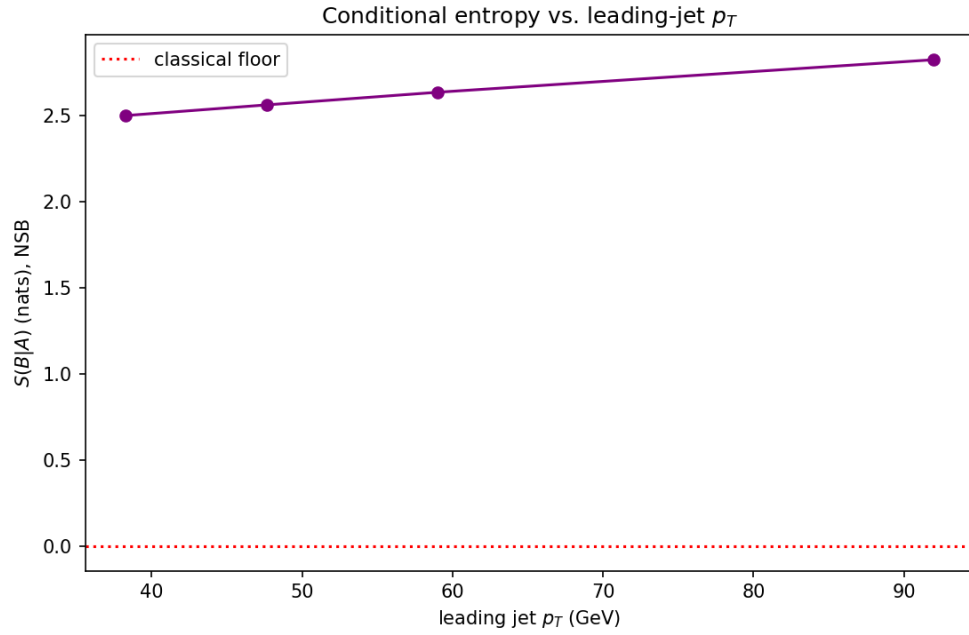


Figure 18: PYTHIA 8: $S(B|A)$ (NSB) vs. leading-jet p_T .

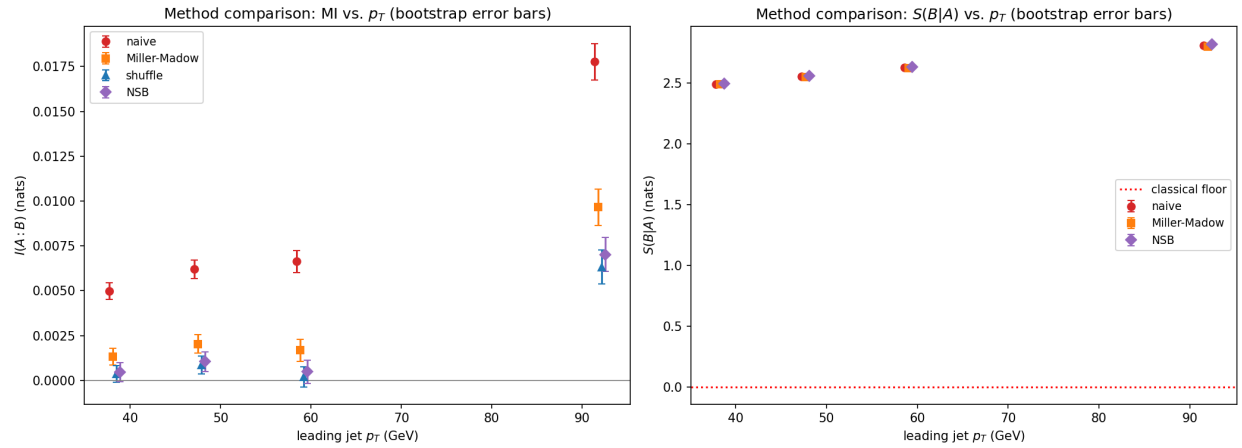


Figure 19: PYTHIA 8: method comparison for $I(A:B)$ (left) and $S(B|A)$ (right) vs. leading-jet p_T , bootstrap error bars throughout.